

FINAL EXPLANATION: BIOLOGY GRADE 11

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 11: REPRODUCTION IN ANIMALS (SUB-STRAND 3.1) FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: How do different animals ensure successful fertilisation and development of offspring, and how can science assist when human reproduction fails? Use what you observed and learned in Lessons 1 to 7 to write your Final Explanation. Answer each prompt in the space below it, using full sentences. Support every claim with evidence from our lessons: the frogs near Kisumu and Lake Victoria wetlands, the hens, the cows, the dissected small mammal, the sperm and ovum slides, the hormone graphs, the animations, the case studies and the debate. Use the correct scientific terms. You may add labelled drawings or flowcharts where they help your explanation. Write in your own words.

Part 1: Frogs, Hens and Cows - Different Strategies, One Goal

Frogs near Kisumu release thousands of jelly-covered eggs into water, hens lay hard-shelled eggs on land, and cows give birth to one live calf. (a) Name the type of fertilisation each animal uses and say where it happens. (b) Classify the hen, the cow, a fish or frog, and a snake as oviparous, viviparous or ovoviviparous, and say how the groups differ. (c) Use evidence from Lesson 1 to explain why frogs need thousands of eggs while a cow produces one calf.

Write your answer here:

Part 2: Structure and Function in Mammals

In Lessons 2 and 3 you dissected a small mammal and studied sperm and ovum slides. (a) Name the parts of the female reproductive system you identified and state the job of the ovaries, the oviduct and the uterus. (b) Describe two structural features of a sperm cell and two of an ovum, and explain how each helps fertilisation. (c) Explain how these structures help a cow, or a human, produce one well-protected offspring rather than thousands of eggs.

Write your answer here:

Part 3: Hormones Control Release and Pregnancy

(a) Using the hormone graphs from Lesson 4, describe how FSH, LH, oestrogen and progesterone change during a 28-day menstrual cycle and what each does. (b) Explain why a woman usually releases one egg per month. Use the idea of feedback in your answer. (c) Describe what happens to hormones and body structures if fertilisation occurs, including the roles of hCG, the placenta, oxytocin and prolactin.

Write your answer here:

Part 4: When Reproduction Fails - IVF, Surrogacy and Sperm Banking

Some people say that IVF babies grow wholly in a test tube. (a) Using what you learned in Lesson 6, explain whether this statement is correct, and describe what happens in IVF and where the baby develops. (b) Choose TWO of the case-study couples (Nakuru, Kisumu, Eldoret, Nairobi or Mombasa) and explain which technology helps each one and why. (c) Give one ethical issue raised in our debate and explain why both parents should share informed decisions.

Write your answer here:

Part 5: Our Final Explanation

Write a connected Final Explanation (one or two paragraphs) that answers the driving question: How do different animals ensure successful fertilisation and development of offspring, and how can science assist when human reproduction fails? Use the frog, hen and cow, the structure of human reproductive systems and gametes, hormones, and IVF or surrogacy or sperm banking in your answer. Then say how your thinking has changed since your first model in Lesson 1.

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Comparing fertilisation strategies	Correctly names external and internal fertilisation and oviparous, viviparous and ovoviviparous types, and explains the number of offspring using the site of fertilisation, losses and parental care, with frog, hen and cow evidence.	Names the types of fertilisation and gives a reasonable explanation of why frogs have more eggs than cows, with limited use of evidence.	Mixes up the types or gives only a statement that frogs lay more eggs, without a reason.
Structure and function	Accurately links reproductive organs and the structures of sperm and ovum (tail, mitochondria, acrosome, protective layers, food reserves) to their functions and to the protection of few offspring.	Names most structures and gives correct functions for some of them.	Lists some structures without functions or with several errors.
Hormonal control	Accurately describes the changes in FSH, LH, oestrogen and progesterone,	Describes the main hormone changes and their roles with minor gaps in	Names a few hormones with unclear or incorrect roles.

	explains single ovulation through feedback, and explains hCG, the placenta, oxytocin and prolactin in pregnancy and birth.	feedback or pregnancy hormones.	
Reproductive biotechnology and ethics	Explains that the baby does not grow wholly in a test tube, matches technologies to specific causes of infertility, and discusses ethical issues and shared decisions clearly.	Explains IVF, surrogacy or sperm banking correctly and mentions an ethical issue.	Describes the technologies vaguely or believes the baby grows wholly in a test tube.
Quality of the Final Explanation	A connected, well-organised explanation that answers the driving question, uses lesson evidence and scientific terms, and reflects on how the thinking has changed.	A mostly complete explanation using some evidence and terms, with limited reflection.	Disconnected or incomplete statements that do not answer the driving question.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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